EXP NO: 13

Implementation of Ping Program.

AIM:

To implement a simple ping program to check if a destination host is reachable and to measure the Round trip time of Packets.

ALGORITHM

i) Start the program, and read the destination host name or IP address from the user.

ii) Create a raw ICMP socket and Echo request containing ICMP type, code, checksum, Identifier, sequence number, Data.

iii) Calculate the checksum of the packet and send the ICMP echo request to the destination

iv) Wait for a reply, then calculate the RTT

v) Display the destination, packet size, sequence number, and RTT.

Code:

```
import socket
import struct
import time
import select
import os
import statistics
ICMP_ECHO_REQUEST = 8
ICMP_ECHO_REPLY = 0
PACKET_SIZE = 64
TIMEOUT = 2
PACKET_COUNT = 4
def checksum(data):
    total = 0
    length = len(data)
    i = 0
    while i < length - 1:
        total += (data[i] << 8) + data[i + 1]
        i += 2
    if length % 2:
        total += data[-1] << 8
    while total >> 16:
        total = (total & 0xFFFF) + (total >> 16)
    return (~total) & 0xFFFF
def create_packet(packet_id, sequence):
    header = struct.pack(
        "!BBHHH",
        ICMP_ECHO_REQUEST, 0, 0,
        packet_id, sequence
```

```
def ping(host, sequence):
    try:
        destination = socket.gethostbyname(host)
        sock = socket.socket(
            socket.AF_INET, socket.SOCK_RAW,
            socket.IPPROTO_ICMP
        )
        sock.settimeout(TIMEOUT)
        packet_id = os.getpid() & 0xFFFF
        packet = create_packet(packet_id, sequence)
        start_time = time.time()
        sock.sendto(packet, (destination, 1))
        while True:
            ready = select.select([sock], [], [], TIMEOUT)
            if not ready[0]:
                print("Request timed out.")
                sock.close()
                return None
            received_packet, address = sock.recvfrom(1024)
            end_time = time.time()
            ip_header_length = (received_packet[0] & 15) * 4
            icmp_header = received_packet[ip_header_length:ip_header_length + 8]
            if len(icmp_header) < 8:
                continue
```

output:

```
Enter ho

Pinging
Sending

Reply f
Reply f
Reply f
Reply

-----
Packet

-----
Minim
Maxi
Aver
```

RESULT:

learn

output:

```
Enter hostname or IP address: 8.8.8.8

Pinging 8.8.8.8 [8.8.8.8]
Sending 4 ICMP packets:

Reply from 8.8.8.8: bytes=56 time=24.35 ms
Reply from 8.8.8.8: bytes=56 time=23.82 ms
Reply from 8.8.8.8: bytes=56 time=25.16 ms
Reply from 8.8.8.8: bytes=56 time=24.02 ms

----- Ping Statistics -----
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

----- Round Trip Time -----
Minimum = 23.82 ms
Maximum = 25.16 ms
Average = 24.34 ms
```

RESULT:

The implementation of Ping Program has been completed successfully